Yingfan (Ivan) Luo

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Education

University of British Columbia Vancouver, BC
Bachelor of Applied Science in Computer Engineering — GPA: 86% *2028*
Relevant Coursework: Computing Systems, Software Construction, Data Structure and Algorithm, Circuit Analysis

Experience

Ruboss Technology Corporation May. 2026 – Aug. 2026
Full Stack Software Engineer *Vancouver, BC*

- Co-developed Leanpub's native iOS app from scratch in **SwiftUI** as one of two developers, building secure authentication, book creation, and voice-input dictation against the **GraphQL** backend via Apollo iOS.
- Built a dual-mode chapter editor supporting both raw **Markua** markup and a visual editor, then shipped a formatting toolbar and keyboard shortcuts shared across the web app and iOS.
- Fixed a production **GraphQL** bug where a non-nullable price field returned `null` and nulled entire book queries, breaking reader pages; shipped a fallback resolver with a regression test in **Ruby on Rails**.
- Built an AI-assisted workflow with **Claude Code**, authoring custom skill files and MCP integrations (Linear, Harvest) and reviewing all generated code before merge.

UBC Rocket - TVR Team Sept. 2025 – Present
Embedded Software Engineer *Vancouver, BC*

- Engineer a high-performance ground control station in **C++** and **Qt**, visualizing 100+ Hz telemetry with 3D attitude visualization and a satellite-view map for real-time position tracking.
- Built the bidirectional radio link: **COBS**-encoded message framing for reliable command/telemetry transfer, an updated **Protobuf** schema, and CSV logging of received packets for post-flight analysis.
- Developed and validated **PID** and flight control loop algorithms with **CTest** test cases that caught regressions across tuning cycles, and added in-station PID tuning presets to accelerate iteration.
- Delivered a **1 kHz** IMU data pipeline with consistent timing by designing non-blocking embedded firmware on **STM32** (SPI-based).
- Developed STM32 firmware for a Qorvo **DW3000** ultra-wideband (UWB) ranging module over UART, integrating it into the vehicle's sensor suite for precise distance/position tracking.
- Developed and tuned **ESC/motor control firmware** and built a **Python** data-collection pipeline to measure force/torque vs. thrust %, enabling performance validation and iteration.

Projects

Unix Shell | *C, Linux*

- Built a Unix shell in **C** with a REPL supporting foreground/background job execution via **fork/execve** and process groups.
- Implemented **job control** (`fg`, `bg`, `kill`, `jobs`) and POSIX signal handlers (`SIGINT`, `SIGTSTP`, `SIGCHLD`), eliminating zombie processes.

Virtual Memory System | *C*

- Implemented virtual-to-physical address translation in C via a **3-level page table** walk, with a **TLB** cache and miss fallback.
- Handled page faults and demand paging with **LRU page replacement** over a limited pool of physical frames.

Technical Skills

Programming Languages: C, C++, Java, JavaScript, TypeScript, Python, Ruby, Swift, SystemVerilog, ARM Assembly
Frameworks: React, Next.js, Remix, Node.js, SwiftUI, Ruby on Rails, GraphQL, Qt, Protobuf
Developer Tools: Git/GitHub, Claude Code, Linear, Xcode, Vercel, Linux, CTest